

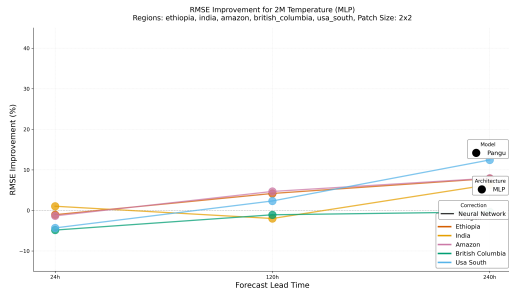
Scratch Work: Tailoring Machine Learning Weather Predictions for Local Impacts

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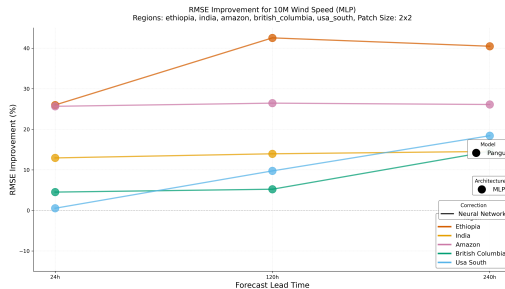
What's new?

- Rewrote databuild to be much faster and allow for using more regions. Early results for Ethiopia
- Added month embeddings to unet and mlp architectures.
- Capable to train on multiple lead times at once using learned lead time embedding.
- Implemented Bayesian hyperparameter optimization and optimized parameters for 2m temperatures 1 week lead time in India.

Do Better with Temperature: 2x2

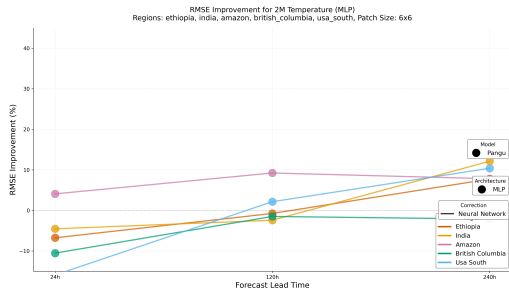


(a) 2m Temperature: Pangu

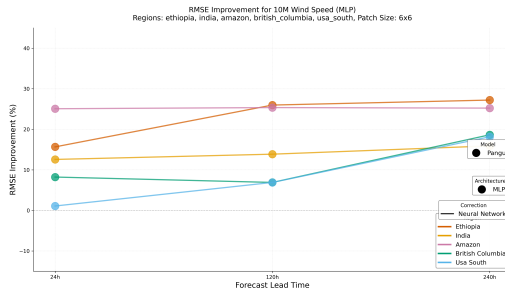


(b) Wind Speed: Pangu

6x6 Region Size

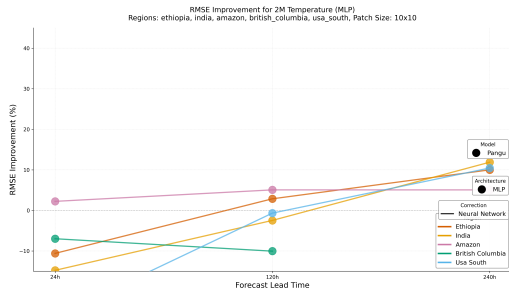


(a) 2m Temperature: Pangu

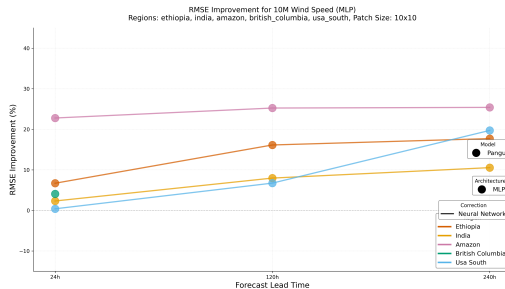


(b) Wind Speed: Pangu

10x10 Region Size



(a) 2m Temperature: Pangu



(b) Wind Speed: Pangu

Summary Table

Region	Lead Time	RMSE	Improvement (%)
Usa South	24h	1.415 (+0.370)	-26.2
<i>Ground Truth: 294.06 (11.74)</i>	120h	2.838 (+0.019)	-0.7
	240h	6.029 (-0.630)	10.5
India	24h	0.895 (+0.132)	-14.8
<i>Ground Truth: 296.15 (5.71)</i>	120h	1.304 (+0.032)	-2.5
	240h	1.789 (-0.213)	11.9
Amazon	24h	0.965 (-0.021)	2.2
<i>Ground Truth: 298.80 (1.50)</i>	120h	1.132 (-0.057)	5.1
	240h	1.343 (-0.068)	5.1
British Columbia	24h	1.482 (+0.104)	-7.0
<i>Ground Truth: 275.86 (13.87)</i>	120h	2.728 (+0.274)	-10.1
	240h	5.844 (-0.336)	5.7
Ethiopia	24h	1.003 (+0.106)	-10.6
<i>Ground Truth: 203.68 (5.30)</i>	120h	1.276 (+0.037)	2.0

Summary Table 2

Region	Ground Truth Mean	Ground Truth 90% CI	RMSE (Orig)	RMSE (C
Usa South	4.20	[1.40, 8.24]	2.44	1.07
India	2.43	[0.83, 4.94]	1.07	0.81
Amazon	1.41	[0.49, 2.50]	0.81	1.33
British Columbia	2.27	[0.70, 4.37]	1.33	1.00
Ethiopia	2.19	[0.48, 5.75]	1.00	